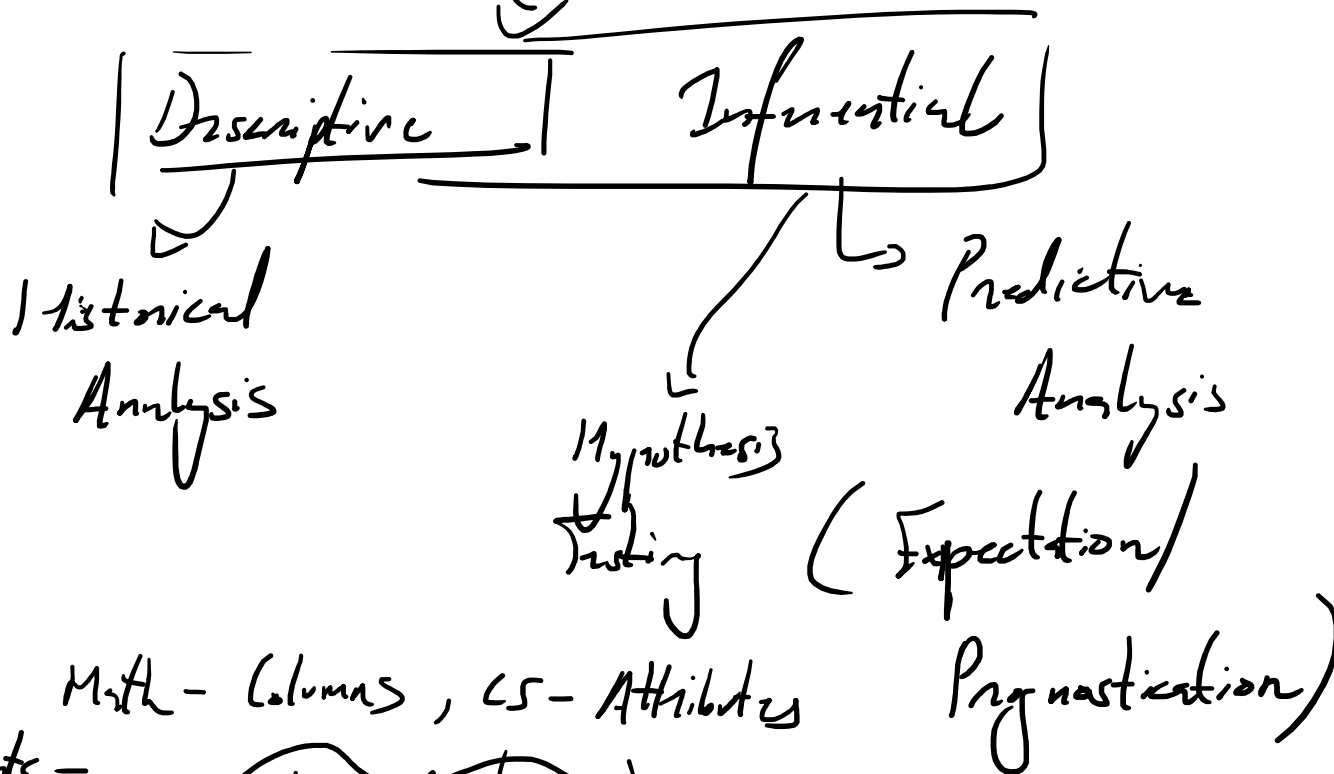


⇒ Intro to ML :

Lin Alg ✓	Calculus ✓	Math
Complex Nes	Statistics ✓	

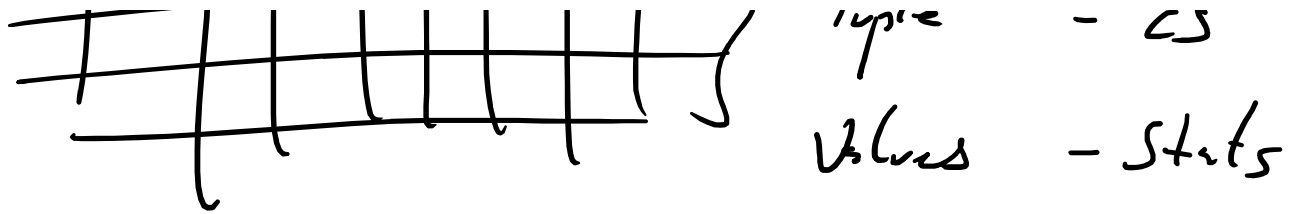


Math - Columns, CS - Attributes

Stats - Features
= S

Rows - Math

Tuple - CS



=> AI learns from patterns.

$\Rightarrow$ Supervised & Unsupervised

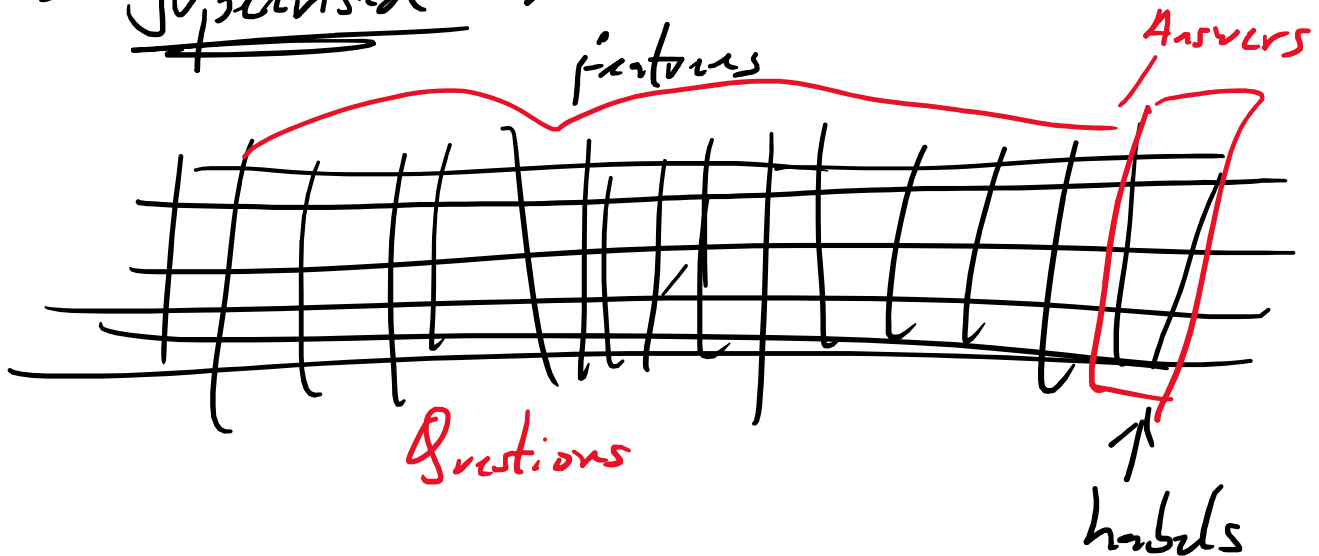
$$MSE \downarrow$$

JEE, SAT, ACT,

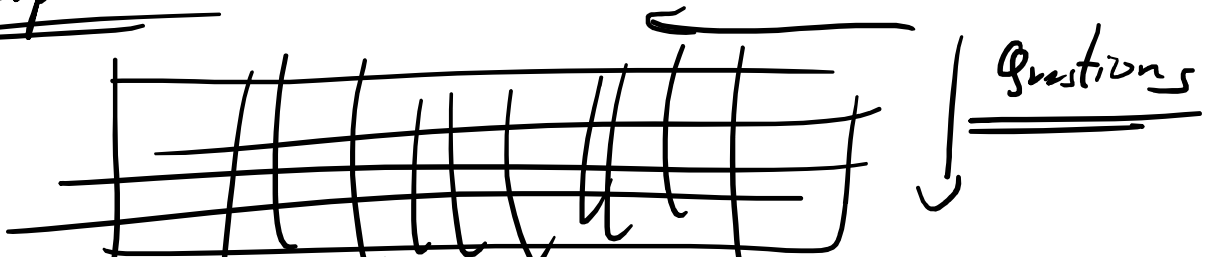
fsE

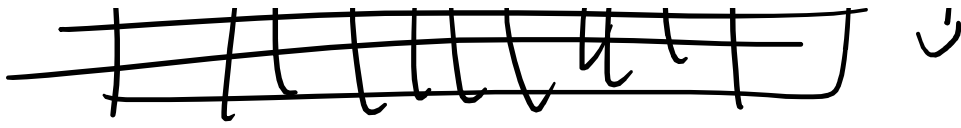
6/25

=> Supervised : Uses calculation
features



→ Unsupervised : User Estimation.





=>

Supervised



Regression

(Prediction of
cont variable)

Classification

(Prediction of
discrete variable)



Parametric & Non Parametric

=> Supervised : Regression

Linear Reg

Multivariate Reg

Polynomial Reg

Ridge Reg

Classification

Logistic Reg

Naive Bayes

K-Nearest Neighbors

Support Vector Mach

Ridge Reg
Lasso Reg

Support Vector Mach
Discriminant Analysis

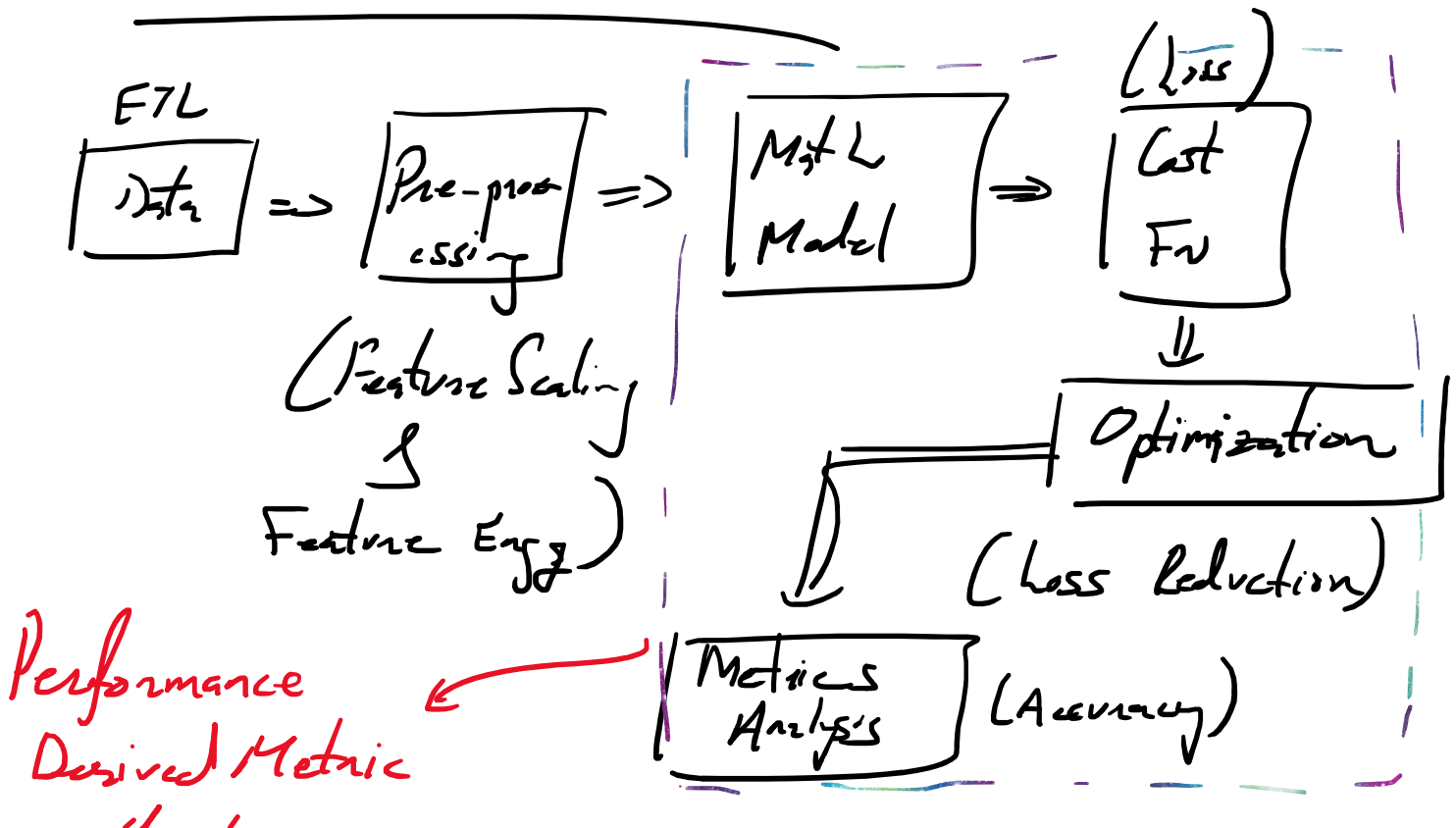
=> Unsupervised Learning: Clustering Algorithms.

Flat-Head
Clustering

Hierarchical
Clustering

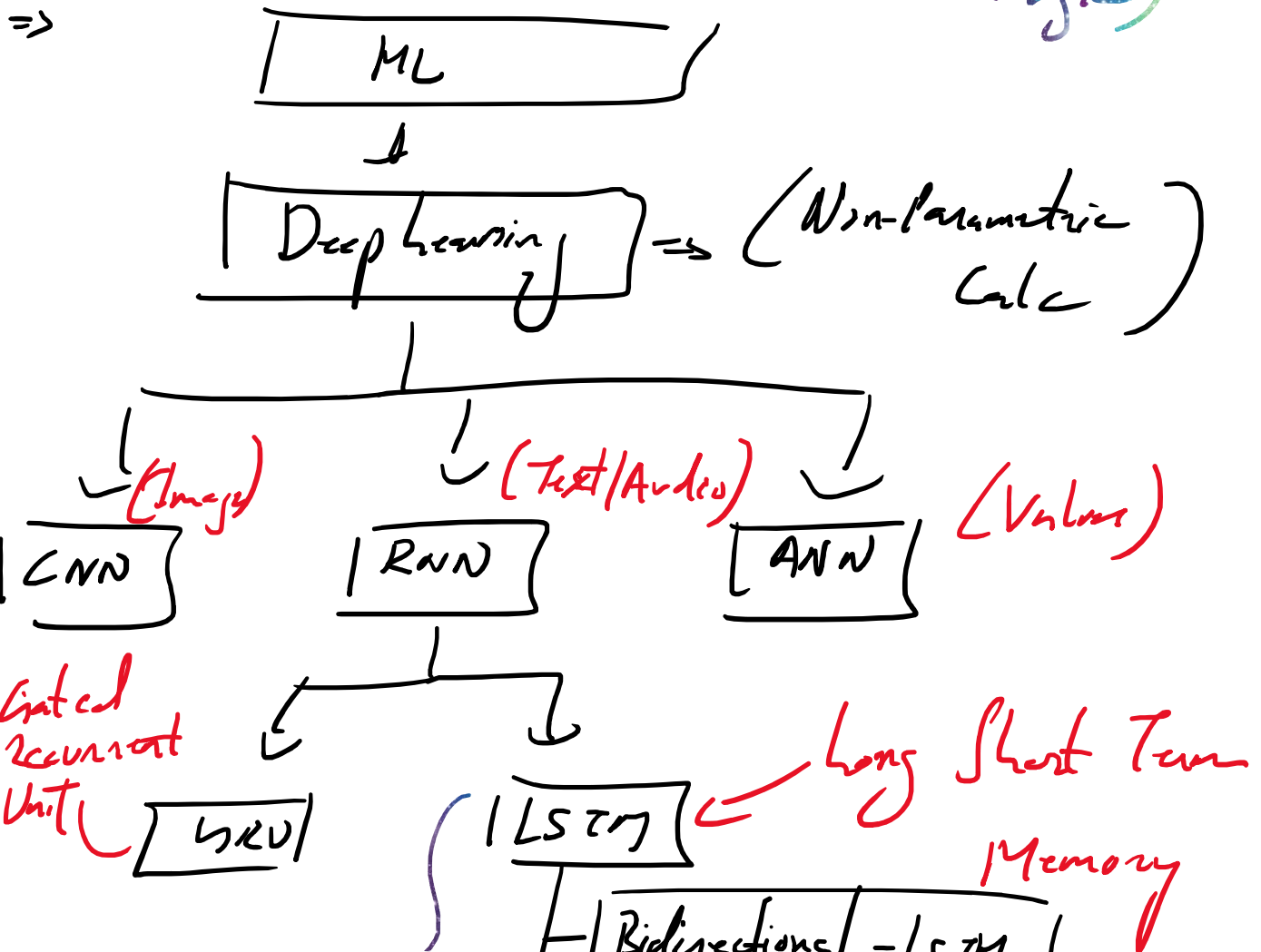
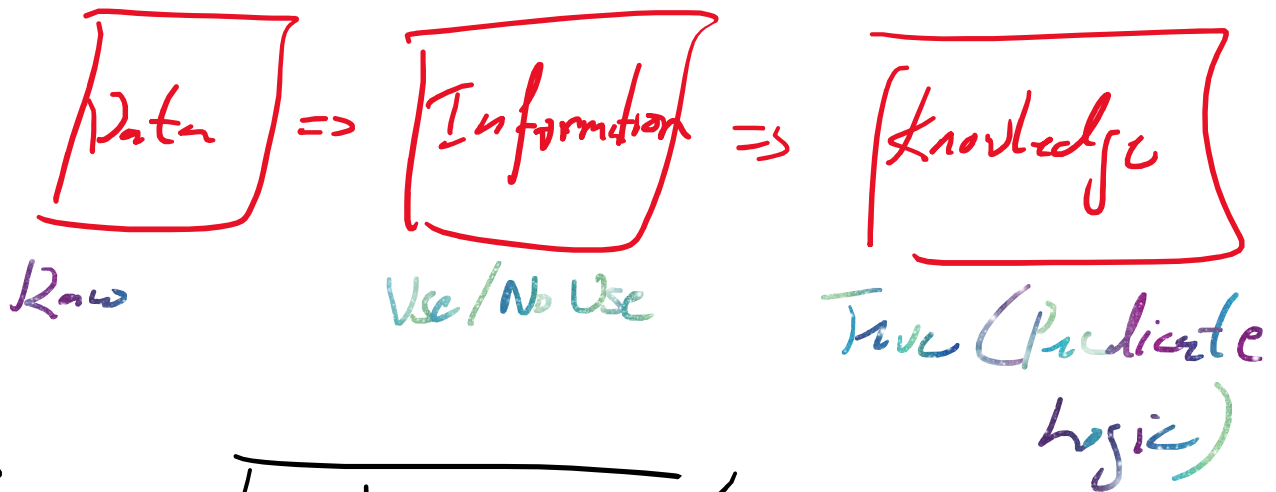
Agglomerative
Clustering

=> How the model works?

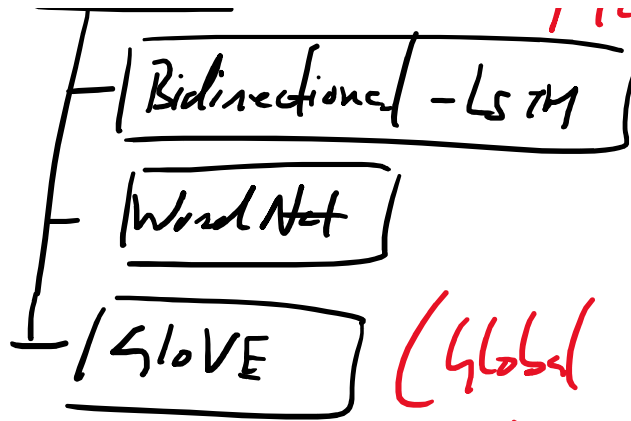


Desired Metric Analysis

=> ETL (Extract Transform Load)



Slow



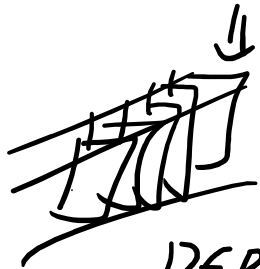
(Global Vector for Embedding)

Attention is all you need

Transformers

Transformer

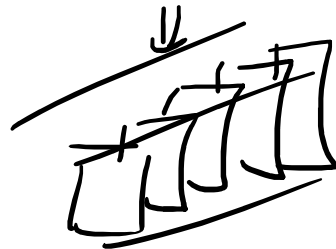
Encoder



BERT

RoBERTa

Decoder



GPT

Generative Pretrained Transformer

=> SCALES:

Nominal Scale : Rank-Based System

Nominal Scale : Rank-Based System

Numerical Scale : Number line

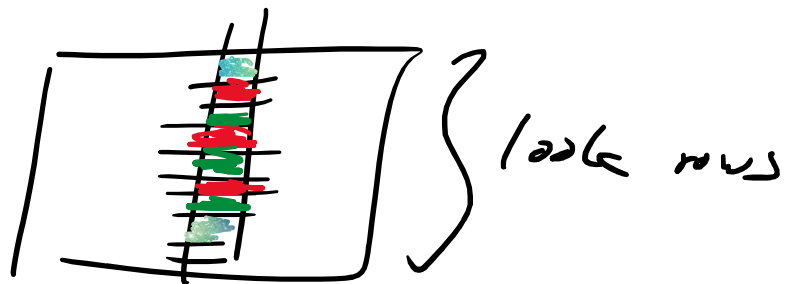
<u>Numerical Variable</u>	&	<u>Categorical Variable</u>
↓		↓
Numbers		Text

(int, float, double, long)

(char, string)

Feature Scaling
⇒ Encoding (Conversion of categorical variable to numerical)

1) Label Encoder:



- i) Extract all unique values (column)
- ii) Assigning discrete values (numerical) to the all unique values.

iii) Replace

	Sex	Smoked
	M	1
	F	0
	M	1

$[F, M]$

$[0, 1]$

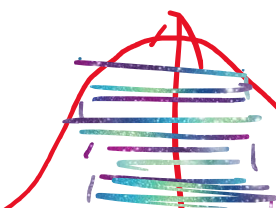
2) One-Hot Encoding: Occurrence

- i) Extract all the unique values.
- ii) Create separate features (columns) for every single unique value.
- iii) In the data add '1' where the value is present, rest all zero.

	Sex	M	F
	M	1	0
	F	0	1
	M	1	0

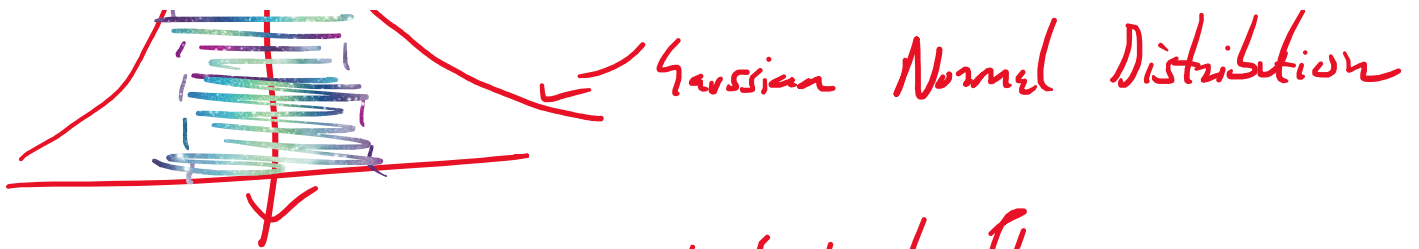
$\begin{matrix} 2 \\ [F, M] \end{matrix}$

$\Rightarrow$ Normalization:

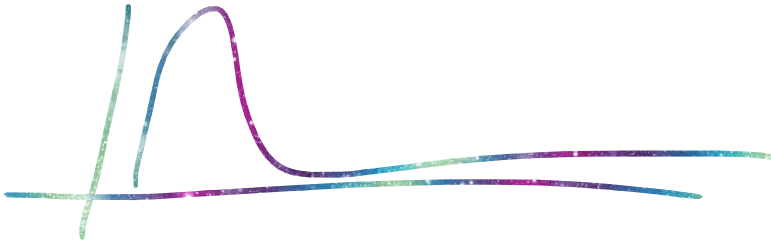


	1.3
	50k
	0.33

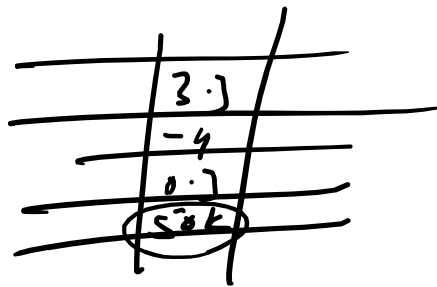
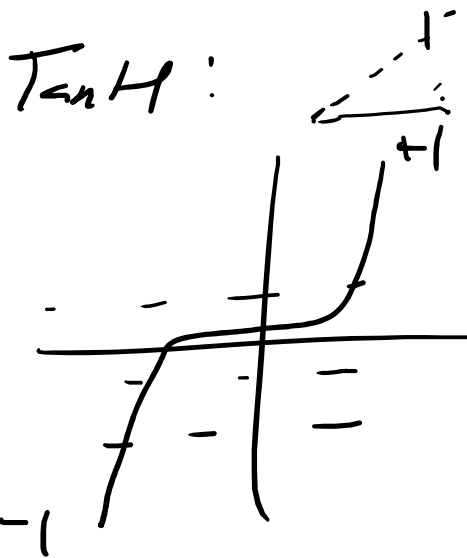
✓ Gaussian Normal Distribution



CLT - Central Limit Theorem



$\Rightarrow$ ReLU : Rectified Linear Unit



$\Rightarrow$ STANDARDIZATION :



